AQ4.1: Activity Questions 1 - Not Graded

This assignment will not be graded and is only for practice.

Level 1

1 point

Choose the set of correct options.

- [ ] If SS is a spanning set of the vector space VV , then $S \cup \{v\}$S ∪ {v} must be a spanning set of VV , for all $v \in V$v ∈ V .
- [ ] Span of an empty set is the zero vector space.
- [ ] If SS is a spanning set of the vector space VV , then $S \setminus \{v\}$S \ {v} must be a spanning set of VV , for all $v \in S$v ∈ S.
- [ ] If SS is a spanning set of the vector space VV , then $S \cup \{v\}$S ∪ {v} may not be a spanning set of VV , for all $v \in V$v ∈ V .

No, the answer is incorrect.

Score: 0

Accepted Answers:

If SS is a spanning set of the vector space VV , then $S \cup \{v\}$S ∪ {v} must be a spanning set of VV , for all $v \in V$v ∈ V .

Span of an empty set is the zero vector space.

Let SS be the set of matrices where $S = \{\begin{bmatrix}1&0\\0&0\end{bmatrix}, \begin{bmatrix}0&1\\0&0\end{bmatrix}, \begin{bmatrix}0&0\\0&1\end{bmatrix}\}$S = {[1000] , [0010] , [0001]}. Consider the following statements:

- Statement P:Statement P: $Span(S)$Span(S) is the vector space consisting of only lower triangular square matrices of order 2.
- Statement Q:Statement Q: $Span(S)$Span(S) is the vector space consisting of only upper triangular square matrices of order 2.
- Statement R:Statement R: $Span(S)$Span(S) is the vector space consisting of all the square matrices of order 2.
- Statement S:Statement S: $Span(S)$Span(S) is the vector space consisting of only scalar matrices of order 2.

Find the number of correct statements.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 1

1 point

If SS is a basis of R^2 R2, then what is the cardinality of SS?

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 2

1 point

Let S = {(1, 2), (1, 3)}S = {(1, 2), (1, 3)} and $span(S) = R^n$ span(S) = Rn. Then find the value of nn.

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 2
1 point
Let SS be the set of matrices where $S = \{\begin{bmatrix}1&0\\0&0\end{bmatrix}, \begin{bmatrix}0&0\\0&1\end{bmatrix}\}$S = {[1000], [0001]}. Consider the following statements:

- Statements P:Statements P: $Span(S)$Span(S) is the vector space consisting of only the Identity matrix of order 2.
- Statements Q:Statements Q: $Span(S)$Span(S) is the vector space consisting of only diagonal matrices of order 2.
- Statements R:Statements R: $Span(S)$Span(S) is the vector space consisting of all the square matrices of order 2.
- Statements S:Statements S: $Span(S)$Span(S) is the vector space consisting of only scalar matrices of order 2.

Find the number of correct statements.

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 1
1 point
1 point
Consider the following set of vectors S = {(1, 1, − 1), (− 1, 1, 1), (0, $\frac{1}{2}$, 0), (0, 1, − 2), (1, 0, − 2)}S = {(1, 1, −1), (−1, 1, 1), (0, 21, 0), (0, 1, −2), (1, 0, −2)}. Choose the set of correct options.

☐ The set { (1, 1, -1), (-1, 1, 1), (0, $\frac{1}{2}$21 ,0)} is not a basis of R^3 R3.

☐ The set {(0, $\frac{1}{2}$21 ,0), (0, 1, -2), (1, 0 ,-2),(1, 0, -2)} is a basis of R^3 R3.

☐ The set {(-1, 1, 1), (0, $\frac{1}{2}$21 ,0), (0, 1, -2)} is a basis of R^3 R3.

☐ None of the above.

No, the answer is incorrect.
Score: 0
Accepted Answers:
The set { (1, 1, -1), (-1, 1, 1), (0, $\frac{1}{2}$21 ,0)} is not a basis of R^3 R3.
The set {(-1, 1, 1), (0, $\frac{1}{2}$21 ,0), (0, 1, -2)} is a basis of R^3 R3.
1 point
Consider a set of vectors S = {(1, 2), (− 1, 2), (3, 1), (− 3, 1)}S = {(1, 2), (−1, 2), (3, 1), (−3, 1)} in R^2 R2. Choose the set of correct options.

☐ The set { (1, 2) ,(-1, 2), (3, 1)} is a basis of R^2 R2.

☐ The set { (-1, 2), (3, 1) } is a basis of R^2 R2.

☐ The set { (3, 1), (-3, 1) } is a basis of R^2 R2.

☐ None of the above.

No, the answer is incorrect.
Score: 0
Accepted Answers:
The set { (-1, 2), (3, 1) } is a basis of R^2 R2.
The set { (3, 1), (-3, 1) } is a basis of R^2 R2.

Level 2

1 point
Choose the set of correct options.

☐ If a subset SS of R^2 R2 contains only two elements then SS is a basis of R^2 R2.

☐ If a subset SS of R^3 R3 contains only one elements then SS can never be a basis of R^3 R3.

☐ If a subset SS of RR contains only one non zero element then SS is a basis of RR.

☐ If S_1 and S_2 S1 and S2 are two bases of R^3 R3 then $S_1 = S_2$ S1 = S2 .

No, the answer is incorrect.
Score: 0
Accepted Answers:
If a subset SS of R^3 R3 contains only one elements then SS can never be a basis of R^3 R3.
If a subset SS of RR contains only one non zero element then SS is a basis of RR.

1 point
Let VV be the subspace of R^3 R3 defined as follows:

$V = \{(x, y, z) \mid x = y - z, \text{ and } x, y, z \in R\}$V = {(x, y, z) ∣ x = y − z, and x, y, z ∈ R}

Choose the set of correct options from the following.

☐ $\{(1, 1, 0), (1, 0, -1)\}${(1, 1, 0), (1, 0, −1)} is a linearly independent set of VV .

☐ $\{(1, 1, 0), (1, 0, -1), (0, 1, 1)\}${(1, 1, 0), (1, 0, −1), (0, 1, 1)} is a linearly independent set of VV .

☐ $\{(0, 1, 1), (1, 0, -1)\}${(0, 1, 1), (1, 0, −1)} is a spanning set of VV .

☐ $\{(1, 1, 0)\}${(1, 1, 0)} is a spanning set of VV .

No, the answer is incorrect.
Score: 0
Accepted Answers:
$\{(1, 1, 0), (1, 0, -1)\}${(1, 1, 0), (1, 0, −1)} is a linearly independent set of VV .
$\{(0, 1, 1), (1, 0, -1)\}${(0, 1, 1), (1, 0, −1)} is a spanning set of VV .

1 point
Let VV and WW be vector spaces which are defined as follows:

$V = \{(x, y) \mid y = mx, \text{ where } m \neq 0 \text{ and } x, y, m \in R\}$V = {(x, y) ∣ y = mx, where m = 0 and x, y, m ∈ R} with usual addition and scalar multiplication as in R^2 R2. $W = \{(x, y) \mid x = 0\}$ with usual addition and scalar multiplication as in R^2.W = {(x, y) ∣ x = 0} with usual addition and scalar multiplication as in R2.

Choose the correct set of options.

- [ ] The set $\{(1, m), (\frac{1}{m}, 1)\}\{(1, \mathrm{m}), (\mathrm{m}1, 1)\}$ is a linearly independent set in V V .
- [ ] The set $\{(1, m), (\frac{1}{m}, 1)\}\{(1, \mathrm{m}), (\mathrm{m}1, 1)\}$ is a spanning set for V V .
- [ ] The set $\{(1, m)\}\{(1, \mathrm{m})\}$ is a linearly independent set in V V .
- [ ] The set $\{(\frac{1}{m}, 1)\}\{(\mathrm{m}1, 1)\}$ is a linearly independent set in V V .
- [ ] The set $\{(0, 1), (0, 2)\}\{(0, 1), (0, 2)\}$ is a linearly independent set in W W
- [ ] The set $\{(0, 1)\}\{(0, 1)\}$ is a linearly independent set in W W .
- [ ] The set $\{(0, 5)\}\{(0, 5)\}$ is a spanning set for W W .

No, the answer is incorrect.
Score: 0
Accepted Answers:
The set $\{(1, m), (\frac{1}{m}, 1)\}\{(1, \mathrm{m}), (\mathrm{m}1, 1)\}$ is a spanning set for V V .
The set $\{(1, m)\}\{(1, \mathrm{m})\}$ is a linearly independent set in V V .
The set $\{(\frac{1}{m}, 1)\}\{(\mathrm{m}1, 1)\}$ is a linearly independent set in V V .
The set $\{(0, 1)\}\{(0, 1)\}$ is a linearly independent set in W W .
The set $\{(0, 5)\}\{(0, 5)\}$ is a spanning set for W W .

Check Answers
Your score is: 0/10